

Sundareswari Thiyagarajan

PhD Applicant in Medical and Multimodal Edge AI

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Research Profile

M.S. researcher in Electrical and Computer Engineering at Inha University specializing in deep learning for speech and biosignals, multimodal architectures, and privacy-aware on-device AI. Published two first-author journal papers in speaker verification and authored research presented at IEEE BigComp 2026. Current work investigates robust sEMG-to-text decoding with a lightweight EMG encoder, an EMG-to-LLM adapter, and quantized Llama 3.2 on NVIDIA Jetson AGX Xavier. Seeking PhD research in biomedical AI, generative and multimodal learning, and efficient edge deployment.

Research Interests

Primary: Biomedical signal AI, sEMG-based silent speech recognition, multimodal learning, privacy-preserving edge intelligence

Methods: Deep neural networks, attention and state-space models, representation learning, generative language models, model quantization

Systems: GPU and embedded deployment, latency-memory benchmarking, reliable on-device inference

Education

M.S. in Electrical and Computer Engineering | Inha University

Feb 2025 - Expected
Feb 2027

Intelligent Embedded Systems Laboratory, Incheon, Republic of Korea | Advisor: Prof. Deok-Hwan Kim | GPA: 4.00/4.50

B.E. in Computer Science and Engineering | Anna University BIT Campus

Aug 2019 - Jun 2023

Tiruchirappalli, Tamil Nadu, India | CGPA: 8.34/10 (3.75/4.5)

Research Experience

Graduate Researcher | Intelligent Embedded Systems Laboratory

Feb 2025 - Present

Department of Electrical and Computer Engineering, Inha University

- Designed deep speaker-verification systems using TDNN, BiLSTM, multi-head attention, state-space modeling, ArcFace/AAM-Softmax, and quality-aware decision support.
- Developed an audio-visual verification pipeline using a ResNet-18 lip encoder, gated fusion, and a shared Conformer; achieved 3.419% EER and 0.342 minDCF on VoxCeleb2 development evaluation.
- Built HySV, a hybrid attention/state-space framework evaluated for verification accuracy and deployment efficiency using EER, minDCF, FLOPs, latency, memory, and real-time factor.
- Benchmark neural and language-model inference across an RTX A6000 workstation and NVIDIA Jetson AGX Xavier, including latency, throughput, memory consumption, and quantized execution.

Current Research Project

SilentEdge-LLM | Privacy-Preserving sEMG Silent Speech Recognition

In Progress

- Developing a multi-channel facial sEMG pipeline combining preprocessing, identity verification, a lightweight EMG encoder, an articulatory-aware representation, and an EMG-to-LLM adapter.
- Targeting local transcript generation with quantized Llama 3.2 3B on Jetson AGX Xavier to keep sensitive biosignals and outputs on-device.
- Studying robustness to cross-session variation, channel loss, limited training data, and electrode-placement changes while measuring edge latency, throughput, and memory.

Selected Publications

- [1] S. Thiyagarajan and D.-H. Kim, "HySV: A Hymba-Inspired Hybrid-Head Framework for Quality-Aware and Deployment-Aware Speaker Verification in Intelligent Embedded Systems." *Electronics*, vol. 15, no. 12, art. 2676, 2026.
- [2] S. Thiyagarajan and D.-H. Kim, "Symmetry and Asymmetry Principles in Deep Speaker Verification Systems: Balancing Robustness and Discrimination through Hybrid Neural Architectures." *Symmetry*, vol. 18, no. 1, art. 121, 2026.
- [3] S. Thiyagarajan and D.-H. Kim, "A Data-Oriented Conceptual Model and Hybrid TDNN-BiLSTM Framework for Context-Aware Speaker Verification in Smart Environments." 2026 IEEE International Conference on Big Data and Smart Computing, 2026.

Selected Publications Continued

[4] S. Thiagarajan, H. Shin, and D.-H. Kim, “A Deep Neural Network Approach for Speaker Verification: Hybrid CNN and Multi-Head Attention with Triplet Loss Framework.” Korean Institute of Information Scientists and Engineers Conference Proceedings, pp. 313-315, 2025.

[5] G. Singh and S. Thiagarajan, “The Multiple Approaches for Drug-Drug Interaction Extraction Using Machine Learning and Transformer-Based Model.” bioRxiv preprint, 2025.

Additional Research and Projects

Stress Detection from Physiological Signals | Machine Learning Project

- Compared LSTM and XGBoost approaches for multimodal physiological-signal classification.

Drug-Drug Interaction Extraction | Biomedical NLP Research

- Investigated classical machine-learning and transformer methods for extracting drug interactions from biomedical text.

Portable Banana Embryo Detection Device | Anna University CSRC Project

- Developed a handheld Arduino and computer-vision prototype through the Student Innovative Project support scheme.

Teaching and Industry Experience

Learning Assistant, Computer Programming Python | Inha University 2026

- Support a 16-week course covering Python fundamentals, data structures, files, object-oriented programming, debugging, and practical exercises.

Laboratory Support, Object-Oriented Programming and C++ | Inha University 2026

- Prepared beginner-to-intermediate programming exercises and guided students during laboratory practice.

Advanced AI Intern | The Korean Academy Jun 2024 - Aug 2024

- Worked on a computer-vision system for identifying flagged vehicles from license-plate imagery using Python and OpenCV.

Java Full-Stack Development Intern | Eazy Bytes Jul 2024

- Completed practical training in Java-based application development and web technologies.

Technical Skills

Programming: Python, C, C++, Java, SQL, HTML, CSS

AI and ML: PyTorch, Hugging Face Transformers, CNN, TDNN, BiLSTM, multi-head attention, state-space models, multimodal fusion

Speech and Biosignals: Speaker verification, audio-visual recognition, sEMG preprocessing, feature extraction, EER, minDCF

Edge Deployment: NVIDIA Jetson AGX Xavier, JetPack, CUDA, TensorRT, Ollama, INT4 quantization, latency and memory benchmarking

Tools: Linux, Git, OpenCV, MATLAB, Arduino IDE, MySQL

Scholarships and Additional Qualifications

Scholarship: Global Vision Graduate Scholarship, Inha University, 2025 - Present

English: IELTS Academic overall band 7.5

Leadership: National Cadet Corps C Certificate, Ministry of Defence, Government of India